

Chapter I: Living organisms

Sources:

- [GCSE Biology - Characteristics of Living Things \(Organisms\) - \[COGNITO\]](#)
- [GCSE Biology - Cell Types and Cell Structure – \[COGNITO\]](#)



Lexicon

To live	
A thing	
Because	
To grow	
To move	
Whole	
Sugar	
i.e.	
Dry	
Larger	
To make	
Same	
To remove	
Waste	
To take	

Definition: A **living organism** is a thing that is alive because it can do things like grow, move, reproduce, adapt to its environment, and do life processes like respiration, reproduction...

I. 7 characteristics of living things: “Mrs Gren”

Movement	The ability for an organism to m_____ (part or whole body)
Respiration	A set of chemical r_____ (i.e. converting sugar to energy)
Sensitivity	Ability to detect and respond to the e_____ (i.e. temperature, hydration...)
Growth	Get larger with increase in dry m_____
Reproduction	Process that m_____ more of the same organism
Excretion	Removal or w_____ produced by the body
Nutrition	Absorbing m_____ from the environment (i.e. food, water, energy)

What you need to know: You need to know the 7 characteristics and be able to explain them

Exercise

Place these items in the table

Cat
Dog
Frog
Human
Ant
Eagle
Tomato
Tulip
Mushroom
Rock
Car
ChatGPT
SmartPhone

Living organism	Not living organism
Plants	
Animals	
Others	

II. What are living organisms made of?

Definition: Cells are the smallest unit of life and the building blocks for all organisms.

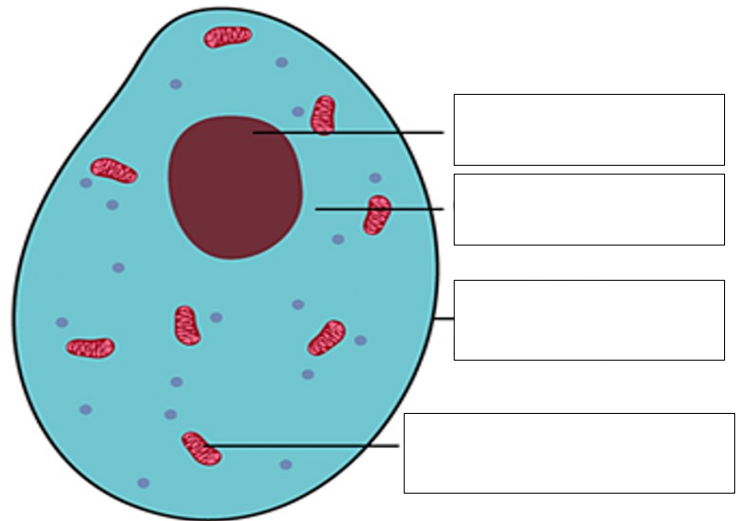
II.1 Animal cell

Nucleus: Contains the genetic material (DNA) of the organism and controls the cell's activities.

Cytoplasm: Liquid that constitutes most of the cell. This is mainly water.

Cell membrane: A flexible outer layer that surrounds the cell and controls what substances can pass in or out.

Mitochondria: Very small "factory" floating in the cytoplasm where energy is released from glucose from food. Where most respiration happens.



II.2 Plant cell

Similar to animal cell with more parts

Cell wall: A rigid outer layer of the cell, which contains cellulose to provide strength and support to the plant.

Vacuole: A space inside the cytoplasm that contains a watery liquid called cell sap. It keeps the cell firm.

Chloroplasts: Structures found in the cells of green parts of plants only (leaves and stems) which contain a green pigment called chlorophyll in which photosynthesis occurs.

